

Lab 4.3 MTConnect Adapter Simulation

3.1 MTConnect Adapter

We have not discussed details of MTConnect adapter yet. However, to see the data stream flow of MTConnect agent, we will simulate MTConnect adapter. The simulation data ([Sensor_log.txt](#)) and simulation adapter ([run_scenario.rb](#)) programmed by ruby can be downloaded from the Github repo [here](#). Please download two files on Raspberry Pi. Please note that two files must be in the same directory in this example. The schematic of the simulation is shown in Figure 8. Raspberry Pi will run MTConnect adapter simulator and data streams are transmitted to the MTConnect agent of laptop. The data log file ('Sensor_log.txt') using Lab1 and Lab2 codes was created by collecting both ADXL345 sensor data (3-axis accelerations), DS18B20 sensor(temperature), and Virtual sensor (humidity). Three sensors have different sampling frequencies. In this scenario, sampling periods are 1 second, 3 seconds, and 3 seconds, respectively. The length of simulation data is 3 minutes, but we will simulate repeating the data continuously. The plots of the data are shown in Figure 9.

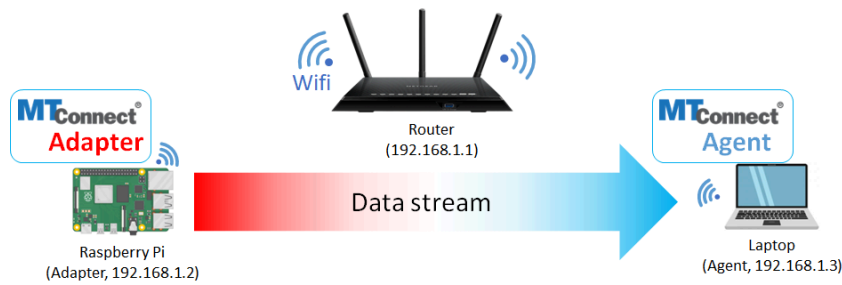


Figure 8 MTConnect schematic for simulation

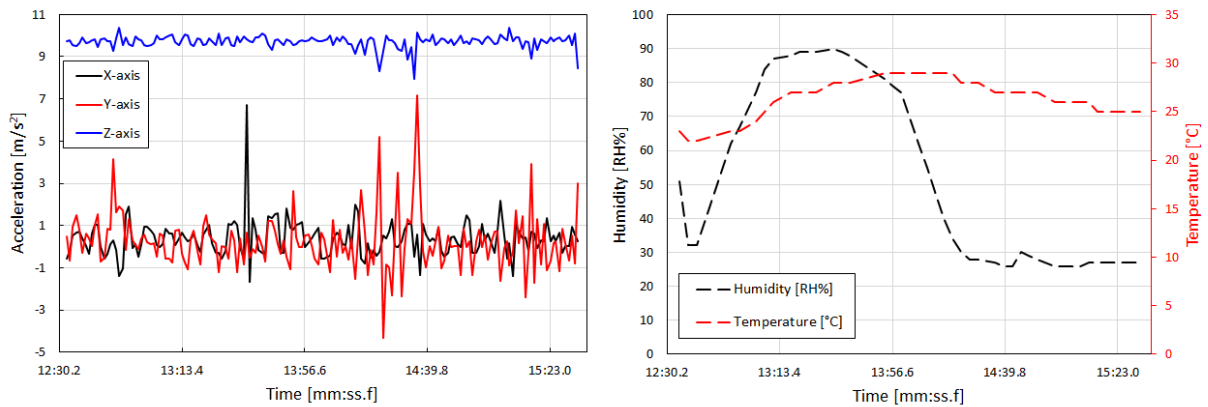


Figure 9 Simulation data ('Sensor_log.txt'): ADXL345 acceleration (left) and DS18B20 & Virtual sensor (right)

First, you need to specify Adapter information in the agent configuration file ('agent.cfg') before running. Please follow the steps below.

1. Open 'agent.cfg' file of laptop using a text editor.
2. Change appropriate Host and Port in 'Adapters { }' according to your network environment.
 - a. Make Port 7878.

b. Host IP address of this example is 192.168.1.2 (your Raspberry Pi IP address must be different). Please make sure of Raspberry Pi's Wifi IP address in the router network.

```
11 Adapters {
12   # Log file has all machines with device name prefixed
13   Adapter1 {
14     Host = 192.168.1.2
15     Port = 7878
16   }
17 }
```

3. Save the file and then close it.
4. Run the agent on laptop.

a. Please make sure that the agent is running before moving on.

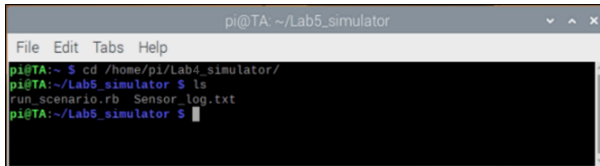
Now, you are ready to run the simulator. Please follow the steps below.

1. Open 'Terminal'.
2. Change directory using 'cd' <file directory where 'run_scenario.rb' and 'Sensor_log.txt' are in> command as below.

Raspberry Pi - Terminal

```
cd /home/pi/Lab4_simulator/
```

- In this example, the file directory is `/home/pi/Lab4_simulator/`.
- Your directory might be different.



```
pi@TA: ~/Lab5_simulator
File Edit Tabs Help
pi@TA:~ $ cd /home/pi/Lab4_simulator/
pi@TA:~/Lab5_simulator $ ls
run_scenario.rb  Sensor_log.txt
pi@TA:~/Lab5_simulator $
```

-
3. Run the simulator with the log file using ruby command as below.

Raspberry Pi - Terminal

```
ruby run_scenario.rb -l -s 192.168.1.2 -p 7878 Sensor_log.txt -v
```

```

pi@noblinmyeong: ~/Lab4_simulator
File Edit Tabs Help
pi@noblinmyeong:~ $ cd Lab4_simulator
pi@noblinmyeong:~/Lab4_simulator $ ruby run_scenario.rb -l -s 192.168.1.116 -p 78
78 Sensor_log.txt -v
Waiting for first connection
Waiting on #<TCPServer:0x0000007f8a0b7840> 7878
Client connected
Waiting on #<TCPServer:0x0000007f8a0b7840> 7878
Received * PING, responding with pong
2026-01-07T16:45:51.997809Z|avail|AVAILABLE
2026-01-07T16:45:51.998665Z|Xacc|-0.5491724|Yacc|0.4707192|Zacc|9.7281968
2026-01-07T16:45:53.008015Z|Xacc|-0.196133|Yacc|-0.6668522|Zacc|9.7674234
2026-01-07T16:45:54.012938Z|Xacc|0.5099458|Yacc|0.9414384|Zacc|9.5320638
2026-01-07T16:45:54.020994Z|temp|23|humd|51
2026-01-07T16:45:55.279725Z|Xacc|0.6668522|Yacc|1.4906107999999998|Zacc|9.492837
1999999999
2026-01-07T16:45:56.282759Z|Xacc|0.7060788|Yacc|0.8237586|Zacc|9.649743599999999
2026-01-07T16:45:57.289161Z|Xacc|0.392266|Yacc|-0.2745862|Zacc|9.924329799999999
2026-01-07T16:45:57.292452Z|temp|22|humd|32
2026-01-07T16:45:58.552351Z|Xacc|0.0392266|Yacc|0.6276256|Zacc|9.649743599999999
2026-01-07T16:45:59.557233Z|Xacc|-0.3530394|Yacc|0.392266|Zacc|9.6889702
2026-01-07T16:46:00.560205Z|Xacc|0.6276256|Yacc|0.0392266|Zacc|9.7674234
2026-01-07T16:46:00.562293Z|temp|22|humd|32
2026-01-07T16:46:01.821154Z|Xacc|1.0198916|Yacc|1.0198916|Zacc|9.80665
2026-01-07T16:46:02.823375Z|Xacc|1.0198916|Yacc|1.5298374|Zacc|9.4536106

```

- The arguments and descriptions are below.
 - ruby : Run Ruby of following *.rb file
 - run_scenario.rb : The MTConnect adapter simulator programed by ruby
 - -l : Loop the file
 - -s : Server IP port to bind to the following IP address
 - 192.168.1.2 : Adapter (here, your PI) IP address
 - -p : Port number of the adapter
 - 7878 : Port number of the adapter
 - Sensor_log.txt : Simulation log filename
 - -v : Verbose output
- Please make sure of your Adapter IP address.
- Please note that to stop (halt) the simulator, press Ctrl + c on 'Terminal'.

4. Check whether it is working or not by 'current' request data to the agent as below.

← → ↻ 🏠 ⚠ Not secure | 192.168.1.3:5000/current

- creationTime: 2022-02-18T08:24:08Z
- sender: DESKTOP-A8SR000
- instanceId: 1645171890
- version: 1.7.0.2
- bufferSize: 131072
- nextSequence: 2190
- firstSequence: 1
- lastSequence: 2189

Device: default; UUID: 000

Device : default

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-02-18T08:24:06.704932Z	Acceleration		Xacc	a1	2187	-1.372931
2022-02-18T08:24:06.704932Z	Acceleration		Yacc	a2	2188	1.9221034
2022-02-18T08:24:06.704932Z	Acceleration		Zacc	a3	2189	10.395049
2022-02-18T08:23:53.627012Z	HumidityRelative	ACTUAL	humd	h1	2155	32
2022-02-18T08:23:53.627012Z	Temperature	ACTUAL	temp	t1	2154	22

Events

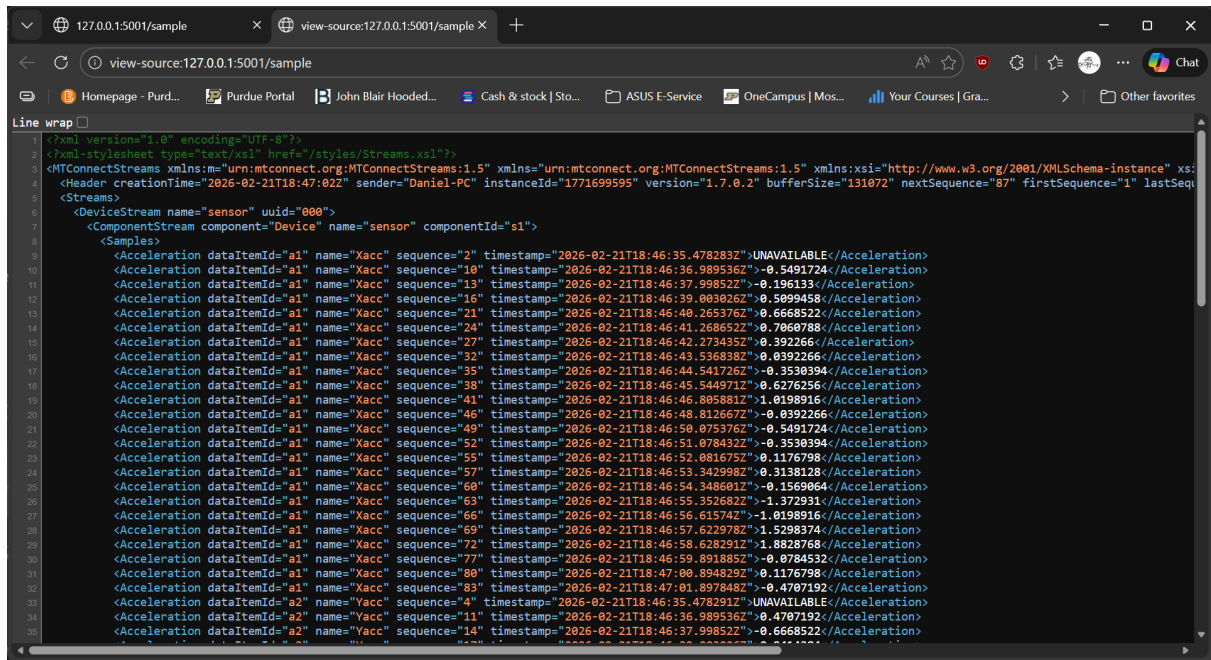
Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-02-18T08:11:48.650723Z	Availability			avail	33	AVAILABLE
2022-02-18T08:11:30.959123Z	AssetChanged			dev_asset_chg	8	UNAVAILABLE
2022-02-18T08:11:30.959114Z	AssetRemoved			dev_asset_rem	7	UNAVAILABLE

Timestamp of data

Data from adapter

Task 3.1

1. Request 'sample' data (laptop: agent, Raspberry Pi: adapter) on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
2. What are your IP addresses (laptop and Raspberry Pi)?



```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="/styles/Streams.xsl"?>
<MTConnectStreams xmlns:m="urn:mtconnect.org:MTConnectStreams:1.5" xmlns="urn:mtconnect.org:MTConnectStreams:1.5" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:
<Header creationTime="2026-02-21T18:47:02Z" sender="Daniel-PC" instanceId="1771699595" version="1.7.0.2" bufferSize="131072" nextSequence="87" firstSequence="1" lastSequ
<Streams>
  <DeviceStream name="sensor" uid="000">
    <ComponentStream component="Device" name="sensor" componentId="s1">
      <Samples>
        <Acceleration dataItemId="a1" name="Xacc" sequence="2" timestamp="2026-02-21T18:46:35.478283Z">UNAVAILABLE</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="10" timestamp="2026-02-21T18:46:36.989536Z">-0.5491724</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="13" timestamp="2026-02-21T18:46:37.998522Z">-0.196133</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="16" timestamp="2026-02-21T18:46:39.003026Z">0.5099458</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="21" timestamp="2026-02-21T18:46:40.265376Z">0.6668522</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="24" timestamp="2026-02-21T18:46:41.260652Z">0.7060708</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="27" timestamp="2026-02-21T18:46:42.273435Z">0.392266</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="32" timestamp="2026-02-21T18:46:43.536838Z">0.0392266</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="35" timestamp="2026-02-21T18:46:44.541726Z">-0.3530394</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="38" timestamp="2026-02-21T18:46:45.544971Z">0.6276256</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="41" timestamp="2026-02-21T18:46:46.805881Z">1.0198916</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="46" timestamp="2026-02-21T18:46:48.812667Z">-0.0392266</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="49" timestamp="2026-02-21T18:46:50.075376Z">-0.5491724</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="52" timestamp="2026-02-21T18:46:51.078432Z">-0.3530394</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="55" timestamp="2026-02-21T18:46:52.081675Z">0.1176798</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="57" timestamp="2026-02-21T18:46:53.342098Z">0.3138128</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="60" timestamp="2026-02-21T18:46:54.348601Z">-0.1560064</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="63" timestamp="2026-02-21T18:46:55.352682Z">-1.372031</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="66" timestamp="2026-02-21T18:46:56.615747Z">-1.0198916</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="69" timestamp="2026-02-21T18:46:57.622978Z">1.5298374</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="72" timestamp="2026-02-21T18:46:58.628291Z">1.8828768</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="77" timestamp="2026-02-21T18:46:59.891885Z">-0.0784532</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="80" timestamp="2026-02-21T18:47:00.894829Z">0.1176798</Acceleration>
        <Acceleration dataItemId="a1" name="Xacc" sequence="83" timestamp="2026-02-21T18:47:01.897848Z">-0.4707192</Acceleration>
        <Acceleration dataItemId="a2" name="Yacc" sequence="4" timestamp="2026-02-21T18:46:35.478291Z">UNAVAILABLE</Acceleration>
        <Acceleration dataItemId="a2" name="Yacc" sequence="11" timestamp="2026-02-21T18:46:36.989536Z">0.4707192</Acceleration>
        <Acceleration dataItemId="a2" name="Yacc" sequence="14" timestamp="2026-02-21T18:46:37.998522Z">0.6668522</Acceleration>
```

Laptop (MTConnect agent): 192.168.50.118

Raspberry Pi (MTconnect adapter-simulator): 192.168.50.190

Task 3.2

1. Repeat Task 3.1 in contrariwise MTConnect configuration as Figure 10, i.e., Raspberry Pi is MTConnect agent and laptop is MTConnect adapter (simulator) in this case.
- The command to run the simulator on 'Command Prompt' is the same with Raspberry Pi
2. Capture 'sample' data on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
 3. Save as 'probe', 'current', 'sample' data from the agent as XML documents (using 'View page source') and then attach all as deliverables in Brightspace.
- Filename
 - Lab4_TASK3.2_Probe.xml
 - Lab4_TASK3.2_Current.xml
 - Lab4_TASK3.2_Sample.xml

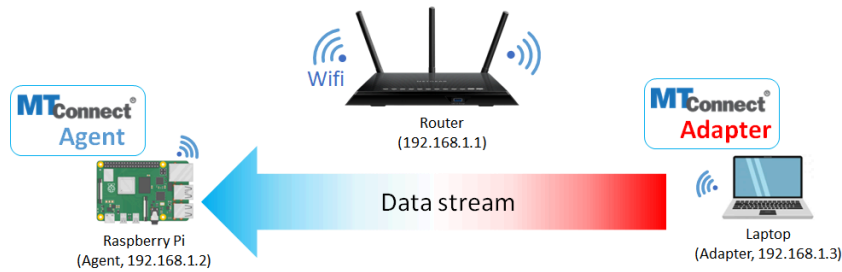


Figure 10 MTConnect schematic for simulation (TASK5)

creationTime: 2026-02-21T18:17:34Z
 sender: robertclaud
 instanceId: 1771697724
 version: 1.8.0.3
 bufferSize: 131072
 nextSequence: 101
 firstSequence: 1
 lastSequence: 381

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-02-21T18:15:24.930233Z	Acceleration		Xacc	a1	7	UNAVAILABLE
2026-02-21T18:15:25.916252Z	Acceleration		Xacc	a1	9	1.0198916
2026-02-21T18:15:27.925212Z	Acceleration		Xacc	a1	14	-0.0392266
2026-02-21T18:15:29.192397Z	Acceleration		Xacc	a1	17	-0.5491724
2026-02-21T18:15:30.196591Z	Acceleration		Xacc	a1	20	-0.3530394
2026-02-21T18:15:31.199606Z	Acceleration		Xacc	a1	23	0.1176798
2026-02-21T18:15:32.464808Z	Acceleration		Xacc	a1	25	0.3138128
2026-02-21T18:15:33.482029Z	Acceleration		Xacc	a1	28	-0.1569064
2026-02-21T18:15:34.499413Z	Acceleration		Xacc	a1	31	-1.372931
2026-02-21T18:15:35.77681Z	Acceleration		Xacc	a1	34	-1.0198916
2026-02-21T18:15:36.780073Z	Acceleration		Xacc	a1	37	1.5298374
2026-02-21T18:15:37.795242Z	Acceleration		Xacc	a1	40	1.8828768
2026-02-21T18:15:39.074807Z	Acceleration		Xacc	a1	45	-0.0784532
2026-02-21T18:15:40.079544Z	Acceleration		Xacc	a1	48	0.1176798
2026-02-21T18:15:41.097334Z	Acceleration		Xacc	a1	51	-0.4707192
2026-02-21T18:15:42.377365Z	Acceleration		Xacc	a1	55	0.1176798
2026-02-21T18:15:43.393379Z	Acceleration		Xacc	a1	58	0.9414384
2026-02-21T18:15:45.673884Z	Acceleration		Xacc	a1	63	0.7453054
2026-02-21T18:15:46.689172Z	Acceleration		Xacc	a1	66	0.588399
2026-02-21T18:15:47.705354Z	Acceleration		Xacc	a1	69	0.0784532
2026-02-21T18:15:48.980687Z	Acceleration		Xacc	a1	74	0
2026-02-21T18:15:49.98785Z	Acceleration		Xacc	a1	77	0.1176798
2026-02-21T18:15:51.007127Z	Acceleration		Xacc	a1	80	0.8629852
2026-02-21T18:15:52.284836Z	Acceleration		Xacc	a1	85	0.6276256
2026-02-21T18:15:54.316388Z	Acceleration		Xacc	a1	89	0.0784532
2026-02-21T18:15:55.609433Z	Acceleration		Xacc	a1	94	0.392266
2026-02-21T18:15:56.626349Z	Acceleration		Xacc	a1	97	0.6668522
2026-02-21T18:15:57.644953Z	Acceleration		Xacc	a1	100	0.4314926
2026-02-21T18:15:24.930025Z	Acceleration		Yacc	a2	6	UNAVAILABLE
2026-02-21T18:15:25.916252Z	Acceleration		Yacc	a2	10	1.0198916
2026-02-21T18:15:26.920473Z	Acceleration		Yacc	a2	12	1.5298374
2026-02-21T18:15:27.925212Z	Acceleration		Yacc	a2	15	-0.7060788
2026-02-21T18:15:29.192397Z	Acceleration		Yacc	a2	18	-0.5099458

When you run ruby simulator on laptop, 'Windows Security Alert' may pop up. Allow network communication for ruby as Figure 11.

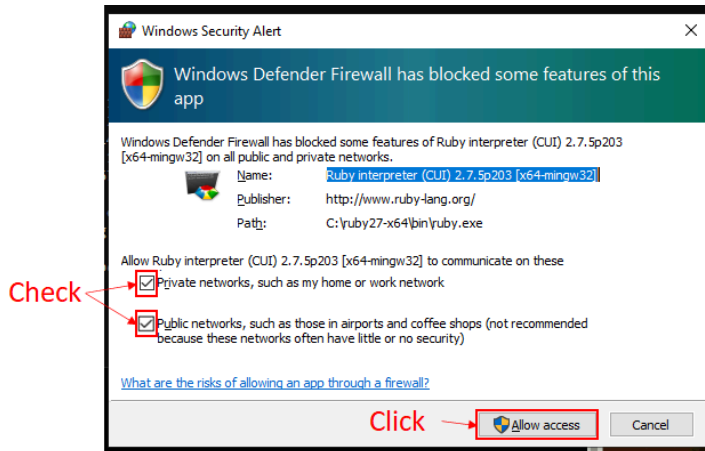
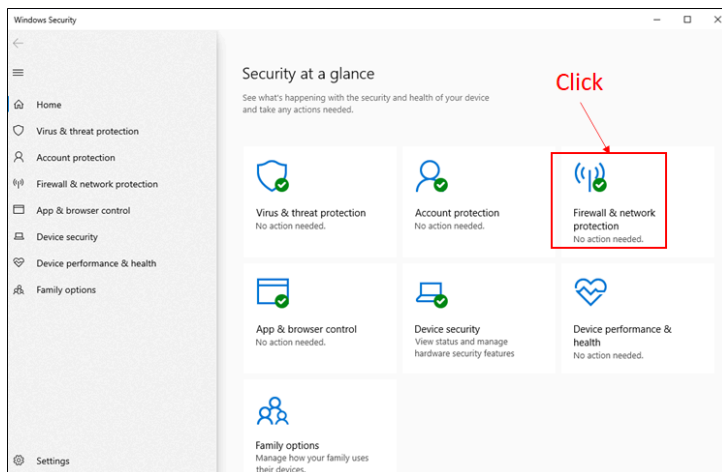


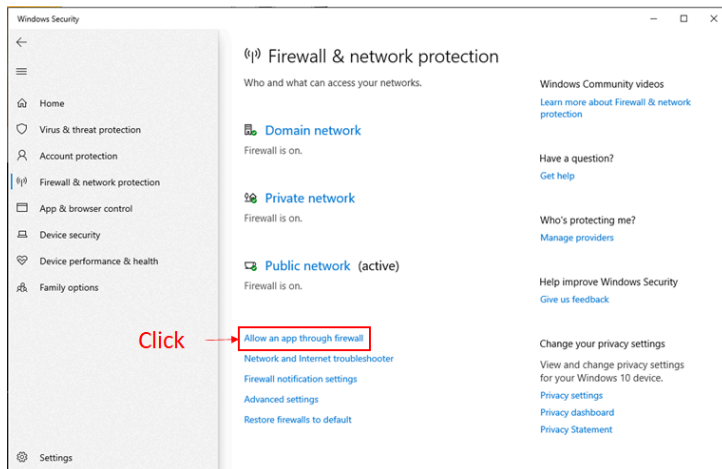
Figure 11 Allow firewall for ruby on Windows

Even if you allow the firewall as above, the simulator (taken a few seconds to run) may not work. If not working, please follow the steps below

1. Open 'Windows Security' and then click 'Firewall & network protection'.



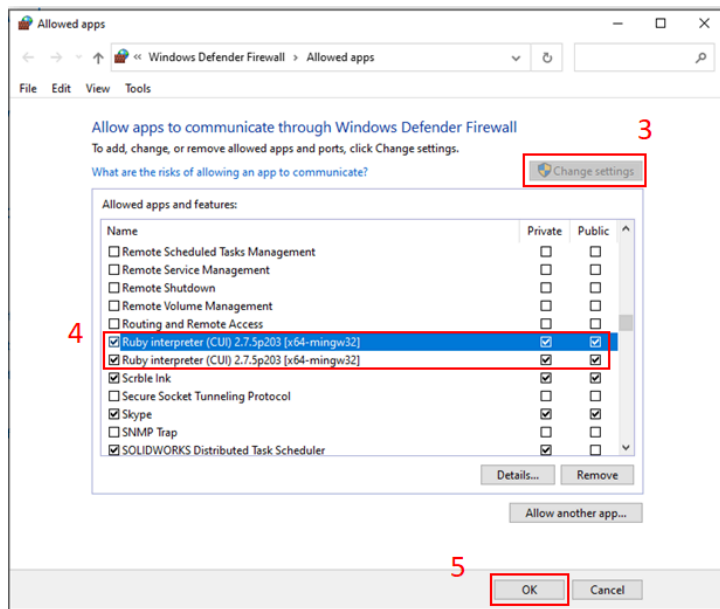
2. Click 'Allow on app through firewall'



3. Click 'Change settings'.

4. Check all the checkboxes related to 'Ruby Interpreter' in Allowed apps and features

5. Click 'OK'.



6. Run the simulator again.

Task 3.3

1. **(DO THIS AT HOME)** Repeat Task 3.1 by running both the agent and adapter (simulation) on laptop as configuration in Figure 12.

- Hint: Adapter IP address should be 127.0.0.1.

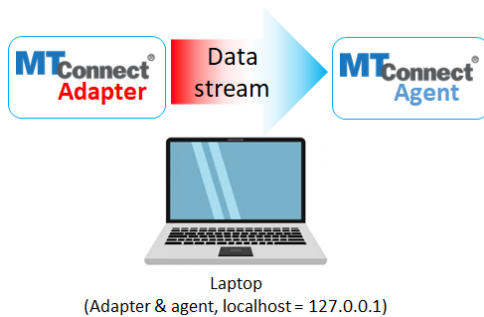


Figure 12 MTConnect schematic for simulation (Deliverable 2)

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xml" href="/styles/Streams.xml"?>
<MTConnectStreams xmlns="urn:mtconnect.org:MTConnectStreams:1.5" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="urn:mtconnect.org:MTConnectStreams:1.5 http://www.w3.org/2001/XMLSchema-instance" creationTime="2026-02-21T18:23:44Z" sender="Daniel-PC" instanceId="1771698159" version="1.7.0.2" bufferSize="131072" nextSequence="197" firstSequence="1" lastSequence="197">
  <Streams>
    <DeviceStream name="sensor" uuid="000">
      <ComponentStream component="Device" name="sensor" componentId="s1">
        <Samples>
          <Acceleration dataItemId="a1" name="Xacc" sequence="194" timestamp="2026-02-21T18:23:44.355134Z">-0.5491724</Acceleration>
          <Acceleration dataItemId="a2" name="Yacc" sequence="195" timestamp="2026-02-21T18:23:44.355134Z">-0.8237586</Acceleration>
          <Acceleration dataItemId="a3" name="Zacc" sequence="196" timestamp="2026-02-21T18:23:44.355134Z">9.9635564</Acceleration>
          <HumidityRelative dataItemId="h1" name="humd" sequence="193" subType="ACTUAL" timestamp="2026-02-21T18:23:43.090092Z">90</HumidityRelative>
          <Temperature dataItemId="t1" name="temp" sequence="192" subType="ACTUAL" timestamp="2026-02-21T18:23:43.090092Z">28</Temperature>
        </Samples>
        <Events>
          <Availability dataItemId="avail" sequence="9" timestamp="2026-02-21T18:22:41.617593Z">AVAILABLE</Availability>
          <AssetChanged dataItemId="s1_asset_chg" sequence="6" timestamp="2026-02-21T18:22:39.808952Z">UNAVAILABLE</AssetChanged>
          <AssetRemoved dataItemId="s1_asset_rem" sequence="5" timestamp="2026-02-21T18:22:39.808949Z">UNAVAILABLE</AssetRemoved>
        </Events>
      </ComponentStream>
    </DeviceStream>
  </Streams>
</MTConnectStreams>
```

Lab4 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab4.

In Lab 4 we learned about the MTConnect protocol to communicate with sensors and devices. We also learned about other communication protocols such as HTTP which enables a web traffic. These two can work together, since MTConnect converts sensor and device outputs into an XML format which can be transferred over HTTP for viewing or into a database. For the lab, we learned how to run an MTConnect agent to monitor the outputs of a device on both our laptops and raspberry pi. We then ran a simulator to simulate the output of a sensor device and were able to view the stream of sensor outputs from an agent either on our pi or laptops.

Q2. What skills did you have to develop to accomplish this project?

I needed to develop more networking skills to accomplish this project. I needed to learn more about network configurations, the IP addresses being used and the ports data was moving through.

Q3. What aspects of this project were the most beneficial for your learning?

The aspects of this project that were the most beneficial for my learning include the prelab overview of the MTConnect and HTTP protocols and the hands on approach where I could see the sensor output through a web browser.

Q4. What challenges did you encounter in completing the project?

In completing the project, I encountered some difficulty installing ruby as I attempted to do parts of the project using a Mac.

Q5. How did you overcome the challenges or remedy the problems encountered?

To remedy the problems encountered, I used the [homebrew](#) package manager to install ruby. After installing ruby using this package manager I also needed to update my PATH variables so that the correct version (3.3.10) was used instead of the system ruby (2.6.3).

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